BRAC UNIVERSITY

Merul Badda, Dhaka, Bangladesh

CSE221 : Algorithms

Quiz 4

Summer 2024

Duration: 20 minutes **Total: 15 marks**

Name:	**ID:**	**Section:**

1. Which one is the **most optimal algorithm** of encoding a single character? In addition, write the time complexity of the algorithm. (5 Marks)

Solution: Huffman Encoding. TC → O(nlogn)

2. Given **4 items** where each item has the **weight [2, 3, 4, 5]** and **profit [1, 2, 5, 6]** associated with it and a bag with **capacity 5**. Simulate a suitable algorithm to select items to put into the bag such that the sum of profits associated with them is the maximum possible. The constraint here is you can either put an item completely into the bag or cannot put it at all. In addition, write the time complexity of your proposed algorithm. (10 marks)

Solution:

0	0	1	1	1	1
0	0	1	2	2	3
0	0	1	2	5	5
0	0	1	2	5	6

TC → O(nw)